



**Generation Interconnection
Impact Study Report
for
Queue Project AE2-301
CRISFIELD 25 KV
2.2 MW Capacity / 5.5 MW Energy**

February, 2020

Table of Contents

1	Preface.....	4
2	General.....	5
2.1	Point of Interconnection	6
2.2	Cost Summary.....	6
3	Transmission Owner Scope of Work.....	7
4	Attachment Facilities	7
5	Direct Connection Cost Estimate.....	7
6	Non-Direct Connection Cost Estimate.....	7
7	Schedule.....	8
8	Interconnection Customer Requirements.....	8
8.1	Direct Connection Requirements	8
8.2	Power Factor Requirement.....	10
8.3	Security Requirements	10
8.4	High Voltage Warning	11
8.5	Additional Operating Requirements	11
9	Revenue Metering and SCADA Requirements	11
9.1	PJM Requirements	11
9.2	DPL Requirements.....	11
10	Network Impacts.....	12
11	Summer Peak Load Flow	12
11.1	Generation Deliverability	12
11.2	Multiple Facility Contingency	12
11.3	Contribution to Previously Identified Overloads.....	12
11.4	Potential Congestion due to Local Energy Deliverability.....	13
11.5	System Reinforcements.....	14
11.6	Flow Gate Details.....	14
11.6.1	Index 1	15
11.7	Affected Systems	17
11.8	Contingency Definitions	17
12	Light Load Analysis	17
12.1	Generation Deliverability	17

12.2	Multiple Facility Contingency	17
12.3	Contribution to Previously Identified Overloads.....	17
12.4	Potential Congestion due to Local Energy Deliverability.....	17
12.5	System Reinforcements.....	18
13	Short Circuit.....	18
14	Stability and Reactive Power Requirement.....	18
15	Attachment 1 – Single Line Diagram.....	19

1 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

2 General

The Interconnection Customer (IC), has proposed a Storage generating facility located in Somerset County, Maryland. The installed facilities will have a total capability of 5.5 MW with 2.2 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is September 22, 2021. This study does not imply a TO commitment to this in-service date.

Queue Number	AE2-301
Project Name	CRISFIELD 25 KV
Interconnection Customer	Able Grid Energy Solutions, Inc.
State	Maryland
County	Somerset
Transmission Owner	DPL
MFO	5.5
MWE	5.5
MWC	2.2
Fuel	Storage
Basecase Study Year	2022

2.1 Point of Interconnection

The IC requested a distribution level Point of Interconnection at the Crisfield 25 kV Substation. AE2-301 will connect to existing feeder MD2212 from the Crisfield Substation T3 Transformer. PJM studied the AE2-301 project as an injection into the Delmarva Power & Light Company system at the Crisfield 69 kV Substation.

2.2 Cost Summary

The AE2-301 project will be responsible for the following costs associated with the physical interconnection of the project:

Description	Total Cost
Attachment Facilities	\$0
Direct Connection Network Upgrade	\$922,092
Non Direct Connection Network Upgrades	\$0
Total Costs	\$922,092

In addition, the AE2-301 project may be responsible for a contribution to the following costs associated with network upgrades:

Description	Total Cost
System Upgrades	\$0

Cost allocations for these upgrades are provided in Section 11.5.

3 Transmission Owner Scope of Work

TO work required to accommodate 5.5 MW of energy storage to existing feeder MD2212 from the Crisfield Substation T3 transformer:

1. A utility operated recloser equipped with the proper relaying and communications will be required.
2. Utility grade primary metering will be required
3. 25 kV (3) 3.3 MW voltage regulator is required.
4. An approximately 0.104 miles of 477 KCM 25kV conductor extension off the existing feeder MD 2212 to the point of interconnection along Crisfield Rd will be required.
5. Generation DTT (Direct Transfer Trip) with fiber installation will be provided to the control center.

4 Attachment Facilities

The total preliminary cost estimate for the Attachment work is given in the table below. These costs do not include CIAC Tax Gross-up.

None.

5 Direct Connection Cost Estimate

The total preliminary cost estimate for the Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
25 kV (3) voltage regulators	\$252,204
Express feeder (0.104 miles)	\$ 79,000
Automatic Circuit Recloser	\$ 85,634
Fiber (0.44 miles)	\$ 53,131
Engineering Overhead	\$ 89,313
Telecommunications	\$150,019
Subtotal Cost	\$709,301
Total Direct Connection Costs with 30% Contingency	\$922,092

6 Non-Direct Connection Cost Estimate

The total preliminary cost estimate for the Non-Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

None.

7 Schedule

The estimated time to complete this work is 18-24 months after receipt of a fully executed interconnection agreement.

8 Interconnection Customer Requirements

The IC is responsible for all design and construction related to activities on their side of the Point of Interconnection. Site preparation, including grading and an access road, as necessary, is assumed to be by the IC. Route selection, line design, and right-of-way acquisition of the direct connect facilities is not included in this report and is the responsibility of the IC.

Protective relaying and metering design and installation must comply with DPL's applicable standards. The IC is also required to provide revenue metering and real-time telemetering data to PJM in conformance with the requirements contained in PJM Manuals M-01 and M-14 and the PJM Tariff.

The IC will be required to make provisions for a voice quality phone ("plain old telephone" or "POT") line within approximately 3 feet of each DPL metering position to facilitate remote interrogation and data collection.

The IC shall provide a protection and interrupting device deemed acceptable by DPL to protect the Facility. The protection and interrupting device shall be located at a mutually agreeable location on the Interconnection Customer side of the Point of Interconnection.

A mutually acceptable means of interrupting and disconnecting the generator with a visible break, able to be tagged and locked out, shall be worked out with DPL Distribution Engineering.

8.1 Direct Connection Requirements

Criteria Limits for Distributed Energy Resource (DER) Connections to the DPL, DPL and Pepco Distribution Systems (less than 69kV):

1. Single Phase Limit

Any DER with a capacity that exceeds 100 kW shall be a balanced three-phase system.

2. Voltage Limits

DER's are permitted to cause a voltage fluctuation of up to 2% at the Point of Interconnection, ½ the band width of any voltage regulator at its terminals, and ½ the net dead band of a switched capacitor bank at its connection point. When a DER is at maximum output, it shall not raise the feeder voltage above the ANSI C84.1 or state limit, whichever is more conservative.

3. Existing Distribution Circuit Capacity Limits

The aggregate limit of large (250 kW and over) generators running in parallel with a single, existing distribution circuit is 0.5 MW on the 4 kV, 3 MW on the 12 kV, 6 MW on the 25 kV, and 10 MW on the 34 kV.

4. Express Circuit Capacity Limits

Distributed generation installations which exceed the criteria limit for an existing circuit require an express circuit.

The maximum generator size for express circuits, depending on transformer capacity, shall be:

- 4 kV 0.5 MW
- 12 – 13.8 kV 10 MW
- 23 – 25 kV 10 MW
- 33.26 – 34.5 kV 15 MW

5. Distribution Power Transformer Limit

The aggregate limit of large (250 kW and over) generator injection to a single distribution transformer of 22.5 MVA nameplate or larger is 10 MW. Transformers with nameplate ratings lower than 22.5 MVA will be given lower ratings on an individual basis. If the transformer rating is significantly greater than 40 MVA it may be possible to consider a greater generation capacity. Adding a new transformer will be considered if there is no availability on any of the existing transformers and space is available in an existing substation. Any proposed transformers would be DPL's standard distribution transformer.

6. Express Circuit Length Limit

If there is no space for an additional transformer at the closest substation, the next closest substation will be considered. The length of an express circuit is limited to 5 miles, or for the sake of the feasibility study, 3.8 straight line miles to the substation. This simplification is used because the feasibility study phase does not allow for the time and resources to examine routes in detail (including existing pole lines, easements, ROW, and environmental issues etc.)

7. When a New Substation is Required

If a distribution express circuit can't be built from an existing substation for a project, it will be necessary to construct a new distribution substation with a standard ring bus design. It will be supplied by extending existing transmission lines. It is the developer's responsibility to verify eligibility of this configuration for solar renewable energy certificates.

All limits, given above in MW, are subject to more detailed study to ensure feasibility.

8.2 Power Factor Requirement

The generators used for this project shall be capable of operating at a power factor (or schedule) specified by DPL in the range of 0.95 leading to 0.95 lagging. It is the responsibility of the developer/customer to obtain equipment that can operate with these requirements while also meeting all applicable requirements of IEEE and UL standards such as, but not limited to, IEEE 1547 and UL 1741.

For this project, operate inverters at a lagging power factor of (0.99) absorbing volt-ampere reactive ("VAR") continuously.

Inverter Requirements (if applicable):

The inverter at the DG location shall have the following capabilities:

- Voltage flicker reduction through dynamic VAR or fixed power factor response
- Ramp rate control
- SCADA communications
- Curtailment or other mitigation ability if high voltage were to occur
- Disturbance Ride through for both Voltage and Frequency
- Ability to receive and respond to a transfer trip signal
- Ability to adjust power factor or VAR based on utility signal
- Ability to Adjust Real Power Output based on utility signal
- Ability to operate on a Volt/VAR schedule
- Ability to maintain a voltage schedule

The inverter(s) shall operate in accordance with both the IEEE 1547 and UL 1741 series of standards that have been approved and use default settings except when specified otherwise by DPL. While inverters should be capable of voltage stabilization through dynamic VAR response and capable of low voltage and system disturbance ride through, neither of these capabilities will be implemented until such time that the IEEE 1547 series of standards are revised and approved to include standards for these capabilities. At such time as these revised standards become available, the generation owner/operator shall cooperate with DPL to implement these capabilities with settings acceptable to DPL. Until such time, the inverters shall operate with a fixed power factor value between 0.95 lead and 0.95 lag as specified by DPL.

8.3 Security Requirements

It is the responsibility of the owner to secure the generator or inverter from any unauthorized access (including physical and remote access) which could alter settings or adversely affect its ability to operate as required. Security measures should include utilizing secure password settings and/or physical locks on cabinet doors.

8.4 High Voltage Warning

Typically, voltage received at the meter from the utility can be up to 105% of nominal (without generation on). Normal operating procedures dictate that voltage at the substation be raised to the higher end of an acceptable bandwidth in order to provide adequate supply to distant customers. It is recommended that transformers with no load taps should be used to adjust secondary voltage to avoid the possibility of inverter trips. Failure to account for this may result in lost energy production.

8.5 Additional Operating Requirements

1. DPL will require the capability to remotely disconnect the generator from the grid by communication from its System Operations facility. This will be accomplished with a line recloser.
2. It is the IC's responsibility to send the data that PJM and DPL requires directly to PJM (or in some cases to DPL directly). The IC will grant permission for PJM to send DPL the following telemetry that the IC sends to PJM: real time MW, MVAR, volts, amperes, generator/status, and interval MWh and MVARh.
3. DPL reserves the right to charge the IC operation and maintenance expenses to maintain the IC attachment facilities, including metering and telecommunications facilities, owned by DPL.

9 Revenue Metering and SCADA Requirements

9.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

9.2 DPL Requirements

DPL interconnection requirements can be found at <https://pjm.com/planning/design-engineering/to-tech-standards/private-dpl.aspx>

10 Network Impacts

The Queue Project AE2-301 was evaluated as a 5.5 MW (Capacity 2.2 MW) injection at Crisfield 69 kV Substation in the DPL area. Project AE2-301 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AE2-301 was studied with a commercial probability of 100%. Potential network impacts were as follows:

11 Summer Peak Load Flow

11.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

11.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

11.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC/D C	MW IMPACT
7728303	232128	PINEY138	138.0	DP&L	232127	LORETT O	138.0	DP&L	1	DPL_P4 - 2_DP58	breaker	158.0	101.58	103.16	AC	2.48

11.4 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPAC T
16346323	232002	CEDAR CK	230.0	DP&L	232013	SILVER RUN	230.0	PJM	1	CKT 23032B	operation	679.0	99.88	100.14	AC	2.12
8106292	232128	PINEY138	138.0	DP&L	232127	LORETT O	138.0	DP&L	1	DPL_P1_2_CKT 13787	operation	158.0	101.66	103.24	AC	2.48

11.5 System Reinforcements

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE2-301	Network Upgrade Number
7728303	1	PINEY138 138.0 kV - LORETTO 138.0 kV Ckt 1	Reinforcement: Reconductor the Piney to Loretto 138 kV line New Ratings: 183/247/247 Cost: \$17,300,000 Time: 32-48 Months Queue Project AE2-301 presently does not receive cost allocation for this upgrade. Note 1: As changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AE2-301 could receive cost allocation. Note 2: Although Queue Project AE2-301 may not have cost responsibility for this upgrade, Queue Project AE2-301 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE2-301 comes into service prior to completion of the upgrade, Queue Project AE2-301 will need an interim study.	\$17.3 M	\$0	N6405
			TOTAL COST	\$17.3 M	\$0	

11.6 Flow Gate Details

The following indices contain additional information about each flowgate presented in the body of the report. For each appendix, a description of the flowgate and its contingency was included for convenience. However, the intent of the appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. Although this information is not used "as is" for cost allocation purposes, it can be used to gauge other generators impact. It should be noted the generator contributions presented in the appendices sections are full contributions, whereas in the body of the report, those contributions take into consideration the commercial probability of each project.

11.6.1 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
7728303	232128	PINEY138	DP&L	232127	LORETTO	DP&L	1	DPL_P4-2_DP58	breaker	158.0	101.58	103.16	AC	2.48

Bus #	Bus	MW Impact
232912	OH NUG1	1.2300
232914	OH NUG3	1.2300
232915	OH NUG4	1.2300
232916	OH NUG5	1.2300
232917	OH NUG6	1.2245
232918	OH NUG7	1.2218
901004	W1-003 E	1.2208
901014	W1-004 E	1.2208
901024	W1-005 E	1.2208
901034	W1-006 E	1.2208
917082	Z2-012 E	3.3179
924681	AB2-120 C	12.0654
924682	AB2-120 E	19.6856
938653	AE1-087 EBAT	1.4200
939151	AE1-145 C1	3.1866
939152	AE1-145 C2	2.1244
939153	AE1-145 E	0.0531
939361	AE1-167 C O1	1.5933
939362	AE1-167 E O1	1.3277
939621	AE1-192 C O1	12.5759
939622	AE1-192 E O1	6.1541
942823	AE2-301 BAT	2.4807
DUCKCREEK	DUCKCREEK	0.0818
NEWTON	NEWTON	0.0982
CHILHOWEE	CHILHOWEE	0.0127
G-007	G-007	0.0588
TVA	TVA	0.1242
PRAIRIE	PRAIRIE	0.1834
O-066	O-066	0.3971
COFFEEN	COFFEEN	0.0375
CARR	CARR	0.0232
EDWARDS	EDWARDS	0.0373
CHEOAH	CHEOAH	0.0355
TILTON	TILTON	0.0447
SANTEETLA	SANTEETLA	0.0104
SMITHLAND	SMITHLAND	0.0147
COTTONWOOD	COTTONWOOD	0.1491
HAMLET	HAMLET	0.0408
BLUEG	BLUEG	0.3557
UNIONPOWER	UNIONPOWER	0.0554
CANNELTON	CANNELTON	0.0216
GIBSON	GIBSON	0.0147

Bus #	Bus	MW Impact
CALDERWOOD	CALDERWOOD	0.0388
FARMERCITY	FARMERCITY	0.0248
TRIMBLE	TRIMBLE	0.0395
RENSSELAER	RENSSELAER	0.0183
CATAWBA	CATAWBA	0.0248

11.7 Affected Systems

None

11.8 Contingency Definitions

Contingency Name	Contingency Definition
DPL_P1_2_CKT 13787	CONTINGENCY 'DPL_P1_2_CKT 13787' OPEN LINE FROM BUS 232132 TO BUS 232130 CIRCUIT 1 /OAK HALL - POCOMOKE 138 END
CKT 23032B	CONTINGENCY 'CKT 23032B' OPEN LINE FROM BUS 232013 TO BUS 232003 CIRCUIT 1 /SILVER RUN - CARTANZA 230 END
DPL_P4-2_DP58	CONTINGENCY 'DPL_P4-2_DP58' /*OAK HALL BUS BREAKER DISCONNECT BRANCH FROM BUS 232132 TO BUS 232130 CKT 1 /*OAK HALL POCOMOKE 138 138 DISCONNECT BRANCH FROM BUS 232132 TO BUS 232131 CKT 1 /*OAK HALL NEW CHURCH 138 138 END

12 Light Load Analysis

12.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

12.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

12.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

12.4 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

12.5 System Reinforcements

None

13 Short Circuit

The following Breakers are overduty:

None

14 Stability and Reactive Power Requirement

Stability Analysis is not required for AE2-301.

15 Attachment 1 – Single Line Diagram

